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The Effectiveness of Gait Retraining on Running Kinematics, Kinetics, Performance, Pain, and Injury in Distance Runners: A Systematic Review With Meta-analysis

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OBJECTIVE: To evaluate the effectiveness of running gait retraining on kinematics, kinetics, performance, pain, and injury in distance runners.

DESIGN: Intervention systematic review with meta-analysis.

LITERATURE SEARCH: Seven electronic databases from inception to March 2021.

TRIAL SELECTION CRITERIA: Randomized controlled trials that (1) evaluated running gait retraining compared to no intervention, usual training, placebo, or standard care and (2) reported biomechanical, physiological, performance, or clinical outcomes.

DATA SYNTHESIS: Random-effects meta-analyses were completed, and the certainty of evidence was judged using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) criteria. We categorized interventions into step rate, non-rearfoot footstrike, impact, ground contact time, and multiparameter subgroups.

RESULTS: We included 19 trials (673 participants). Moderate-certainty evidence indicated step rate gait retraining increased step rate (SMD 1.03 [95% confidence interval {CI}: 0.63, 1.44]; number of trials (N): 4; I^2: 0%) and reduced average vertical loading rate (SMD -0.57 [95% CI, -1.05 to -0.09], N: 3; I^2: 0%). Low-certainty evidence indicated non-rearfoot footstrike retraining increased knee flexion at initial contact (SMD 0.74 [95% CI, 0.11 to 1.37]; N: 2; I^2: 0%), but did not alter running economy (SMD 0.21 [95% CI, -1.11 to 1.52]; N: 3; I^2: 19%).). Low-certainty evidence indicated multiparameter retraining did not alter running economy (SMD 0.32 [-0.39, 1.02]; N: 3; I^2: 19%) or performance (SMD 0.14 [95% CI, -4.87 to 4.58]; N: 2; I^2: 18%). Insufficient trials reported on pain outcomes. Two trials demonstrated reduced 1-year injury incidence following gait retraining.

CONCLUSIONS: Gait retraining interventions altered step rate and knee kinematics, lowered vertical loading rates, and did not affect running performance. *J Orthop Sports Phys Ther 2022;52(4):192-206. Epub 05 Feb 2022. doi:10.2519/jospt.2022.10585*

KEY WORDS: *running, rehabilitation, biomechanics, gait analysis, injury prevention, gait retraining*

Despite numerous health benefits of running, the risk of sustaining a running-related injury is high.[29,46] The reported incidence of running-related injuries ranges from 3% to 85%,[29,46] with 2.5-33 injuries per 1000 hours of running.[47]

Based on a theoretical link between running biomechanics and injury, researchers and clinicians have implemented interventions aimed at retraining running gait for injury treatment[16,19] or prevention.[8] Gait retraining uses feedback strategies, such as real-time visual feedback of impact loading[11,12] or joint kinematics,[43,53] verbal feedback from a therapist,[19] or audible feedback from an external device (eg, metronome).[23] Biomechanical changes in response to feedback may facilitate a more efficient running technique[13,21] and reduce running-related injuries.[8]

In 2015, there was limited evidence of the effects of gait retraining on clinical outcomes and an absence of randomized controlled trials (RCTs).[4] However, international experts continued to advocate for gait retraining as a treatment and injury prevention intervention.[4] Since 2015, trials investigating the effects of gait retraining on biomechanical,[17,50] pain,[6,19] injury,[8] and performance[42] outcomes have been published. Therefore, we aimed to review and synthesize evidence from RCTs that investigated the

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effect of gait retraining on running kinematics, kinetics, performance, pain, and injury in distance runners.

METHODS

Protocol and Registration

This systematic review was completed following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement[33] and registered with the PROSPERO (International Prospective Register of Systematic Reviews) database (registration number CRD42020153925) and the Open Science Framework (https://doi.org/10.17605/OSF.IO/NGPV6) before the completion of the search.

Search Strategy

We searched the following databases from inception to March 18, 2021: MEDLINE, EMBASE, CINAHL, Web of Science (limited to following topics: sports sciences, orthopedics, physiology, rehabilitation, engineering, biomedicine, multidisciplinary sciences, and biophysics), SPORTDiscus, PEDro, and Cochrane Central Register of Controlled Trials (CENTRAL). Trials were limited to *English* and *Humans*. The complete search syntax for each database is available in **APPENDIX A** (available at www.jospt.org). We used backward (hand search) and forward citation tracking (via Scopus) to identify any remaining publications. We contacted content experts to review the list of trials identified by our search and provide suggestions for additional eligible trials that we may have missed.

Selection Criteria

We included RCTs published in English in a peer-reviewed scientific journal that (1) compared gait retraining (eg, real-time feedback via haptic, visual, or auditory methods) with no intervention, usual training, placebo, or standard care (eg, advice and education, and orthotics); (2) measured at least 1 biomechanical, physiological, performance, or clinical outcome at baseline and at least 1 follow-up time point; and (3) included adult, able-bodied distance runners (≥ 18 years of age). We excluded trials if (1) they solely analyzed walking, sprinting, or a change of direction running; (2) they only varied surface, incline, and velocity or did not provide active instructions to modify gait; (3) they varied footwear (eg, the addition of minimal shoes); or (4) they only examined immediate effects of modifying gait (ie, within the same session). We defined "gait retraining" as an intervention designed to modify or alter running gait patterns and include both an *acquisition phase* (eg, real-time feedback via haptic, visual, or auditory feedback) and a *transfer phase* (eg, a period of supervised or independent practice of modified gait).[14]

Trial Selection

Two authors independently screened the titles and abstracts using a web-based systematic review management platform (Covidence, Melbourne, Australia). Full texts for the remaining trials were reviewed independently by 2 authors. Any discrepancies were resolved by consensus.

Data Extraction

Data extraction from all included trials was carried out by 1 author, cross-checked for accuracy by a second author, and imported into Microsoft Excel (version 2103, 2021). If tabulated data were not presented within the trials, a web-based plot digitizer[41] was used to digitize figures into numeric values. We extracted publication details (author, year, and country), participant demographics (age, sex, injury, and running history), details of the retraining intervention (duration, follow-up, and retraining method), and trial outcomes. The outcomes extracted were spatiotemporal (step rate, step length, vertical oscillation, and ground contact time), kinematic (joint angles of the hip, knee, and ankle) and kinetic variables (vertical loading rates and impact peaks, peak segmental acceleration, and joint reaction forces), performance measures (timed run, running economy, and perceived exertion), pain (validated outcome measures and visual analog scale), and injury (injury incidence, absence from training, and adverse events). Outcome data were extracted for the longest follow-up time point when a trial reported outcomes at multiple time points. We considered longer-term effects more practically meaningful than short-term effects that may not have a lasting impact on runners. The corresponding authors of included trials were contacted to clarify any missing or insufficient data.

Risk of Bias

Two authors assessed the risk of bias of each included trial using the Cochrane Risk of Bias Tool.[25] We assessed *selection bias*, *performance bias*, *detection bias*, *attrition bias*, *reporting bias*, and *other bias*, which covered any additional trial-specific biases that were not covered by the other domains (eg, financial interests in the gait retraining method being investigated). We included 2 additional domains: *awareness of timing of sampling* that considered potential bias related to participants being aware of the exact timing of data collection (eg, running on a force plate) and *compliance with the intervention* that considered potential compliance bias concerning the intervention. Any disparities were resolved via consensus. We explicitly judged all criteria using low risk, high risk, and unclear risk of bias.

Data Synthesis and Analysis

R Studio (version 1.4.11063, Boston, United States) was used to perform random-effects meta-analysis estimated using the restricted maximum-likelihood approach with the Hartung-Knapp correction.[28] We pooled data for all outcomes that demonstrated sufficient homogeneity across at least 2 trials using equivalent gait retraining interventions (see example code in **APPENDIX B**, available at www.jospt.org).

We categorized interventions into 5 distinct subgroups based on the primary gait retraining strategy employed: step rate (targeting an increase in step rate),

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impact (targeting a reduction in landing impacts), non-rearfoot footstrike (transitioning runners from a rearfoot to a midfoot or a forefoot footstrike), ground contact time (targeting a reduction in ground contact time), and multiparameter (using a combination of technical cues; eg, Pose Method). Where trials had a primary gait retraining focus but also included other minor retraining strategies (eg, increase step rate by 7.5%-10% and, if necessary, run softer or adopt a non-rearfoot footstrike pattern[19]), they were categorized based on the primary focus rather than considered to be a multiparameter intervention.

Follow-up means and standard deviations of outcomes were extracted from included trials (or obtained from the authors, if not presented) and used to calculate standardized mean differences (SMDs) with 95% confidence intervals (CIs). *P* values less than .05 were considered statistically significant. When 2 or more control groups were included in trials, we selected the group that most represented usual care (ie, active control). SMDs were described as trivial (< 0.20), small (0.20-0.59), moderate (0.60-1.19), large (1.20-1.99), and very large (≥ 2.00).[27] Pooled mean differences (MDs) with 95% CI were also calculated whenever trials used the same measurement scale. These MDs were reported in-text to compliment the SMDs in the meta-analysis forest plots and assist with interpreting the practical meaning of findings. Statistical heterogeneity was assessed using I^2 statistic and considered low (I^2 < 25%), moderate (I^2 = 25%-49%), or high (I^2 > 50%).[26] Publication bias was not assessed because there were no outcomes with data from 10 or more trials.

Certainty of Evidence

We judged the certainty of evidence as high, moderate, low, or very low using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system.[3] Certainty was downgraded 1 level for each of the following limitations: Total sample size < 400 participants (imprecision); high statistical heterogeneity I^2 > 50% (inconsistency); and more than 50% of trials in the meta-analysis with high risk of bias in more than 1 bias domain. Judgments were performed by 1 author and cross-checked by another author.

RESULTS

Search Results

The search resulted in 14 079 unique titles, and abstracts after duplicates were removed. The screening process and reasons for exclusion are detailed in **FIGURE 1**. A total of 19 trials from 16 independent cohorts met the inclusion criteria. Forward citation searching was completed on March 22, 2021 and resulted in 382 trials but did not identify any further trials meeting the inclusion criteria. No additional trials were identified from backward citation searches or through expert consultation.

Trial Characteristics

Trial characteristics are presented in **TABLE 1**. A total of 673 participants (347 intervention; 326 control) from the 16 independent trial populations were included. Sixteen trials recruited recreational runners; 3 trials[13,21,22] included well-trained or sub-elite runners. Most trials recruited healthy participants; 1 trial[30] included runners with prehypertension, and 2 trials[19,43] included runners with symptomatic patellofemoral pain syndrome. Gait retraining sessions

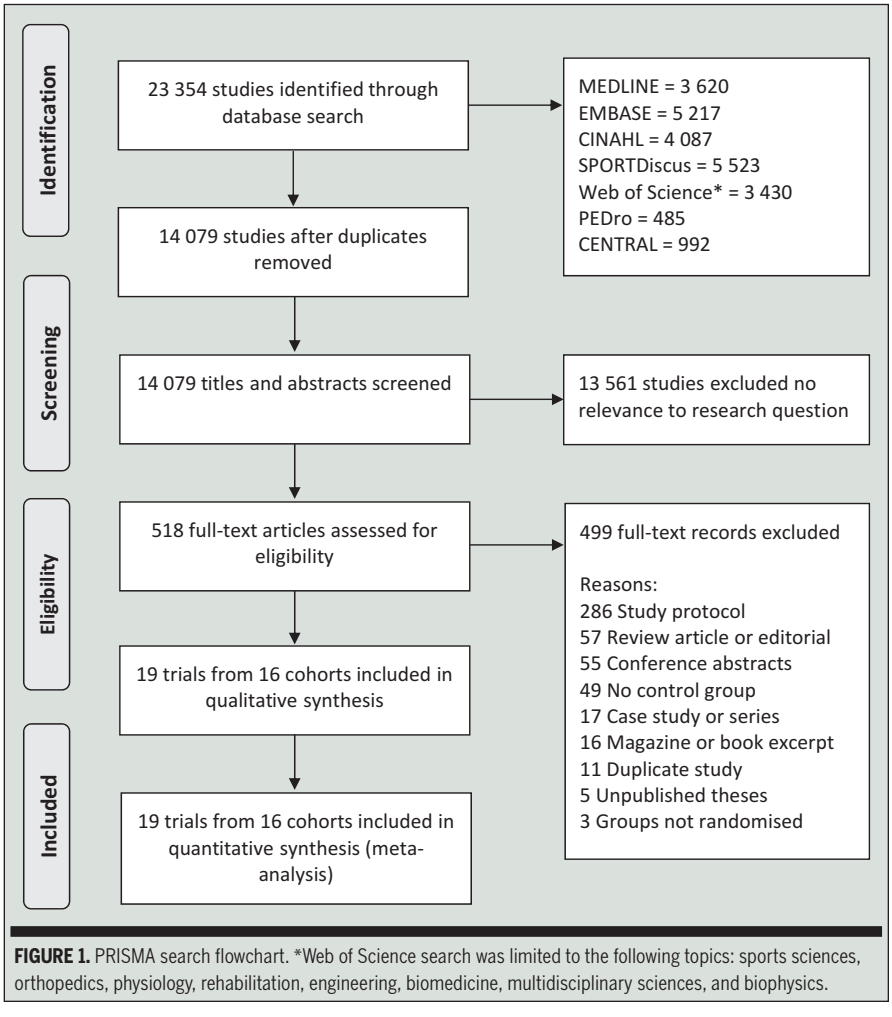

FIGURE 1. PRISMA search flowchart. *Web of Science search was limited to the following topics: sports sciences, orthopedics, physiology, rehabilitation, engineering, biomedicine, multidisciplinary sciences, and biophysics.

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TABLE 1 Study Characteristics

Study	Trial Population	Outcome Measure	Intervention Subgroup	Protocol	Control	Follow-up
Baumgartner et al.[5]	38 recreational runners (16 F, 22 M) Aged 18 yr or older Running at least 15 miles/wk Preferred stride rate below 85 strides/min	Kinematics	Step rate	6 wk of independent running with visual feedback via Garmin device to increase step rate by 10%	No feedback	6 wk
Chan et al.[8]	320 novice runners (162 F, 158 M) Aged 18-50 yr Running over 8 km/wk AVLR > 70 BW/s during usual running speed	Kinetics Injury	Impact	2 wk (8 × 30-min sessions) of visual and verbal faded feedback to lower vertical impact peak	No feedback	2 wk (52 wk for injury outcomes)
Clansey et al.[9]	22 recreational runners (M) Running over 30 km/wk Average peak tibial acceleration greater than 9 g	Kinematics Kinetics Economy	Impact	3 wk (6 × 20-min sessions) of visual and verbal feedback to reduce peak tibial acceleration (tibial shock)	No feedback	7 wk
Craighead et al.[10]	18 recreational runners (sex NR) Aged 18 yr or older Running a minimum of 3 d/wk at least 5 km/d	Kinematics Performance	Non-rearfoot footstrike	8 wk (5 × 60-min sessions) using verbal feedback instructing MFS	General education material unrelated to gait retraining	9 wk
Dallam et al.[13]	16 sub-elite triathletes (M) Able to 10 km at a moderate altitude of 2000 m in under 42 min or less	Kinematics Performance	Multiparameter	12 wk (1 × 3-h initial session + 12 × 60-min sessions) of instruction of Pose Method	Maintain usual training	12 wk
Dunn et al.[17]	20 recreational runners (10 F, 10 M) Aged 18-45 yr	Kinematics Performance	Multiparameter	1 wk (3 × 2-h sessions) of instruction of Pose Method	Maintain usual training	7 wk
Ekizos et al.[18]	32 runners (14 F, 18 M) Running at least 10 km (2 sessions/wk) Habitual RFS	Kinematics Performance	Non-rearfoot footstrike	14 wk (14 × 30-min sessions) of feedback using pressure insoles to transition for RFS to FFS/MFS	Short term intervention or control group that maintained usual training (2 control groups)	15 wk
Esculier et al.[19,20]	69 runners with PFP (43 F, 26 M) Aged 18-45 yr Running at least 15 km/wk PFP over 3 mo	Kinematics Kinetics Pain	Step rate	8 wk (5 × 10-min sessions) of verbal feedback to increase SR by 7.5%-10% (plus if necessary, run softer or with non-rearfoot footstrike)	Education on load management or education and strengthening (2 control groups)	20 wk
Fletcher et al.[21]	16 sub-elite runners (M) Aged 18-30 yr 10 km personal best of 37-45 min Habitual RFS	Kinematics Performance	Multiparameter	1 wk (7 × 60-min sessions) of instruction of Pose Method	Traditional running drills	1.5 wk
Gilgen-Ammann et al.[22]	30 well-trained runners (12 F, 18 M)	Kinematics Performance	Ground contact time	4 wk (8 sessions) of visual feedback to decrease ground contact time	No feedback or own training routine (2 control groups)	6 wk
Kumar et al.[30]	17 runners with prehypertension and mean body mass index under 35 (8 F, 9 M) Mean blood pressure in range (130-150/80-100 mm Hg)	Kinetics	Multiparameter	8 wk (4 × 120-min workshops) of Chi Running instruction	Group-based training or self-directed training (2 control groups)	10 wk
Letafatkar et al.[32]	49 runners (M) Running at least 8 km/wk for at least 2 yr AVLR > 70 BW/s during running at 8 and 12 km/hr	Kinematics Kinetics Injury	Multiparameter	8 wk (24 × 15- to 30-min sessions) of verbal feedback, including instructions to avoid RFS, run softer, keep knees apart and kneecaps pointing ahead	Conditioning program without feedback or placebo exercises (2 control groups)	52 wk

Table continues on page 196.

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TABLE 1 Study Characteristics (continued)

Study	Trial Population	Outcome Measure	Intervention Subgroup	Protocol	Control	Follow-up
Messier et al.[35]	22 novice runners (F) College undergraduate students At least 1 mechanical flaw while running (eg, vertical oscillation, excessive trunk movement, under- or overstride)	Kinematics Performance	Multiparameter	5 wk (15 × 20-min sessions) of real-time verbal and visual feedback	No feedback	5 wk
Roper et al.[42,43]	16 runners with PFP (11 F, 5 M) Self-reported RFS Self-reported as RFS runners Mild-to-moderate chronic, running-related knee pain	Kinematics Performance Pain	Non-rearfoot footstrike	2 wk (8 × 15- to 30-min sessions) of faded visual (mirror) and verbal feedback to transition from RFS to FFS	No verbal feedback	6 wk
Wang et al.[48]	24 recreational runners (M) Running at least 15 km/wk	Kinematics Kinetics	Step rate	12 wk (36 × 30-min sessions) of independent running with audible feedback via a mobile-based metronome application to increase step rate by 7.5%	No verbal feedback	12 wk
Willy et al.[50,52]	30 recreational runners (14 F, 16 M) Aged 18-35 yr Running at least 10 km/wk IVLR > 85 BW/s during running at 3.3 m/s	Kinematics Kinetics	Step rate	8 sessions of independent running with visual feedback via Garmin device (only runs 1-3, 5, 7) to increase step rate by 7.5%	Maintain usual training	6 wk

Abbreviations: AVLR, average vertical loading rate; BW, body weight; F, female; FFS, forefoot strike; IVLR, instantaneous vertical loading rate; M, male; MFS, midfoot strike; NR, not reported; PFP, patellofemoral pain; RFS, rearfoot strike.

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ranged from 1 to 36 sessions with a typical duration of 2-8 weeks. Biomechanical, performance, and clinical outcomes were collected from participants ranging from 1.5 to 52 weeks; the most common follow-up was 6-8 weeks. Both trials that analyzed injury outcomes[8,32] reported injury incidence after 1 year.

Intervention Subgroups

The retraining intervention type varied considerably between trials. Four trials[5,19,48,50] used a step rate intervention that targeted a step rate increase of between 7.5% and 10% from baseline. Three trials[10,18,43] used an intervention that aimed to help runners transition from a rearfoot footstrike toward either a midfoot or a forefoot footstrike. Two trials[8,9] used impact interventions, where participants were provided with real-time visual feedback of their landing impacts. One trial[22] used an intervention that targeted a reduction in ground contact time. The remaining 6 trials[13,17,21,30,32,35] used multiparameter interventions. These interventions used a combination of feedback strategies, such as instructing an increase in step rate, cueing a softer landing, or encouraging a midfoot or forefoot footstrike. Some multiparameter interventions included the use of popular running technique approaches such as Pose Method[13,21] or Chi Running.[30]

Risk of Bias

Risk of bias assessment results are described in **APPENDIX C** (available at www.jospt.org), including the author's judgements for (1) each risk of bias item and (2) percentages across all included trials. Selection bias was considered low risk in 8 trials[5,8,19,30,32,43,48,50] that reported using random sequence generation and 5 trials[8,19,30,32,43] that reported concealing allocation and considered high risk in 1 trial[48] using an open random allocation method, with all remaining trials unclear risk. Performance bias was considered high risk for all trials as it was not possible to blind participants to gait retraining. Detection bias was high risk in 1 trial[5] that reported investigators were not blinded. In comparison, the risk was low in 1 trial[19] due to clearly described blinding of outcome assessors and unclear in the remainder of trials. Three trials[9,18,30] were at high risk of attrition bias due to high participant dropout, 1 trial[13] was unclear, and all remaining trials were considered low risk due to low dropout. Participants were aware of data collection timing in 8 trials[9,17,18,21,30,32,43,48] and judged as high risk due to the potential Hawthorne effect (ie, changing behavior while being observed). Compliance with interventions was considered low risk in 9 trials,[5,9,19,22,30,32,43,50] with the remainder of trials unclear risk due to a lack of compliance information. Other potential biases were detected in 4 trials[10,13,17,21]

where a coauthor or a contributor was the developer of the gait retraining method employed. Finally, an additional trial[30] was considered high risk due to not including a between-groups factor in the statistical analysis.

Spatiotemporal Outcomes

Twelve trials[5,10,13,17-19,21,22,35,43,48,50] reported at least 1 spatiotemporal outcome following running gait retraining. SMD pooled results and grades of evidence are presented in **FIGURE 2** and **TABLE 2**, respectively, whereas corresponding MDs are reported in this subsection text.

Step Rate Nine trials[5,10,18,19,21,22,43,48,50] reported step rate outcomes. There was moderate-certainty evidence from 4 trials[5,19,48,50] that step rate interventions cause a moderate significant increase in step rate (MD 10.0 steps/min [95% CI, 4.8 to 15.3]). There was low-certainty evidence from 3 trials[10,18,43] that non-rearfoot footstrike interventions have trivial nonsignificant effects on step rate (MD 1.7 steps/min [95% CI, -2.3 to 5.8]).

Stride Length Seven trials[10,13,21,35,43,48,50] reported stride length outcomes. There was low-certainty evidence from 2 trials that step rate[48,50] and non-rearfoot footstrike[10,43] interventions have no significant effect on stride length (MD -0.08 m [95% CI, -0.46 to 0.29]) and -0.03 m [95% CI, -0.85 to 0.78], respectively). There was very low–certainty evidence from 3 trials[13,21,35] that multiparameter interventions have a nonsignificant effect on stride length (MD -0.08 m [95% CI, -0.46 to 0.30]). The pooled SMD for all 3 subgroups indicated a small-to-moderate effect but with high imprecision due to large CIs (**FIGURE 2**).

Vertical Oscillation There was very low–certainty evidence from 4 trials[13,17,21,35] that multiparameter interventions have trivial nonsignificant effects on vertical oscillation (MD 0.00 m [95% CI, -0.03 to 0.02]).

Ground Contact Time Two trials[18,22] reported ground contact time outcomes: 1 from the non-rearfoot footstrike subgroup and 1 from the ground contact time subgroup. Due to different intervention subgroups, a pooled analysis was not possible. The trial using a non-rearfoot footstrike intervention[18] reported a very large and significant decrease in ground contact time (MD -38 ms [95% CI, -51 to -25]). Only trivial nonsignificant changes were observed in the trial using the ground contact time intervention (MD 0 ms [95% CI, -15 to 15]).[22]

Kinematic Outcomes

Nine trials[9,10,17,19,32,35,43,48,50] reported kinematic outcomes following running gait retraining. SMD pooled results and grades of evidence are presented in **FIGURE 3** and **TABLE 2**, respectively, whereas corresponding MDs are reported in this subsection text.

Ankle Dorsiflexion Angle at Initial Contact There was very low–certainty evidence from 2 trials[10,43] that non-rearfoot footstrike interventions cause a moderate nonsignificant decrease in ankle dorsiflexion angle (resulting in more ankle plantarflexion) at initial contact (MD 10.4° [95% CI, -116.5 to 95.6]).

Knee Flexion Angle at Initial Contact There was low-certainty evidence from 2 trials[10,43] that non-rearfoot footstrike interventions cause a moderate significant increase in knee flexion angle at initial contact (MD 4.2° [95% CI, 2.3 to 6.1]).

Peak Knee Flexion Angle During Stance There was low-certainty evidence from 2 trials[10,43] that non-rearfoot footstrike interventions have a trivial nonsignificant effect on peak knee flexion angle (MD -1.3° [95% CI, -23.2 to 20.5]). There was low-certainty evidence from 2 trials[17,35] that multiparameter interventions result in a small nonsignificant increase in peak knee flexion angle during stance (MD 2.1° [95% CI, -44.6 to 48.9]).

Peak Hip Adduction Angle During Stance There was moderate-certainty evidence from 2 trials[19,50] that step rate interventions cause a small nonsignificant decrease in peak hip adduction angle during stance (MD -1.3° [95% CI, -6.6 to 4.1]).

Footstrike Angle at Initial Contact There was moderate-certainty evidence from 2 trials[48,50] that step rate interventions cause a moderate nonsignificant decrease in footstrike angle at initial contact (ie, tending toward a forefoot footstrike) (MD -3.5° [95% CI, -5.6 to -1.4]).

Kinetic Outcomes

Eight trials[8,9,19,30,32,43,48,50] reported kinetic outcomes following running gait retraining. SMD pooled results and grades of evidence are presented in **FIGURE 4** and **TABLE 2**, respectively, whereas corresponding MDs are reported in this subsection text.

Average Vertical Loading Rate There was moderate-certainty evidence from 3 trials[19,48,50] that step rate interventions produce a small significant reduction in average vertical loading rate (MD -11.0 bodyweight/second (BW/s) [95% CI, -21.1 to -0.8]). There was low-certainty evidence from 2 trials[8,9] that impact interventions cause a moderate nonsignificant reduction in average vertical loading rate (MD -14.4 BW/s [95% CI, -95.4 to 66.6]). There was low-certainty evidence from 2 trials[30,32] that multiparameter interventions cause a small nonsignificant decrease in average vertical loading rate (MD -5.9 BW/s [95% CI, -19.8 to 8.0]).

Instantaneous Vertical Loading Rate There was moderate-certainty evidence from 2 trials[8,9] that impact interventions cause a moderate nonsignificant reduction in instantaneous vertical loading rate (MD -21.7 BW/s [95% CI, -22.0 to -21.4]). There was also moderate-certainty evidence from 2 trials[48,50] that step rate interventions result in a moderate nonsignificant decrease in instantaneous vertical loading rate (MD -17.3 BW/s [95% CI, -45.0 to 10.5]).

Peak Tibial Acceleration Two trials,[9,32] from the impact and multiparameter subgroups, reported on peak tibial acceleration. The multiparameter intervention[32] caused a large significant decrease in peak tibial acceleration (MD -2.7 g [95% CI, -4.2 to -1.2]); the impact intervention[9] demonstrated a moderate nonsignificant

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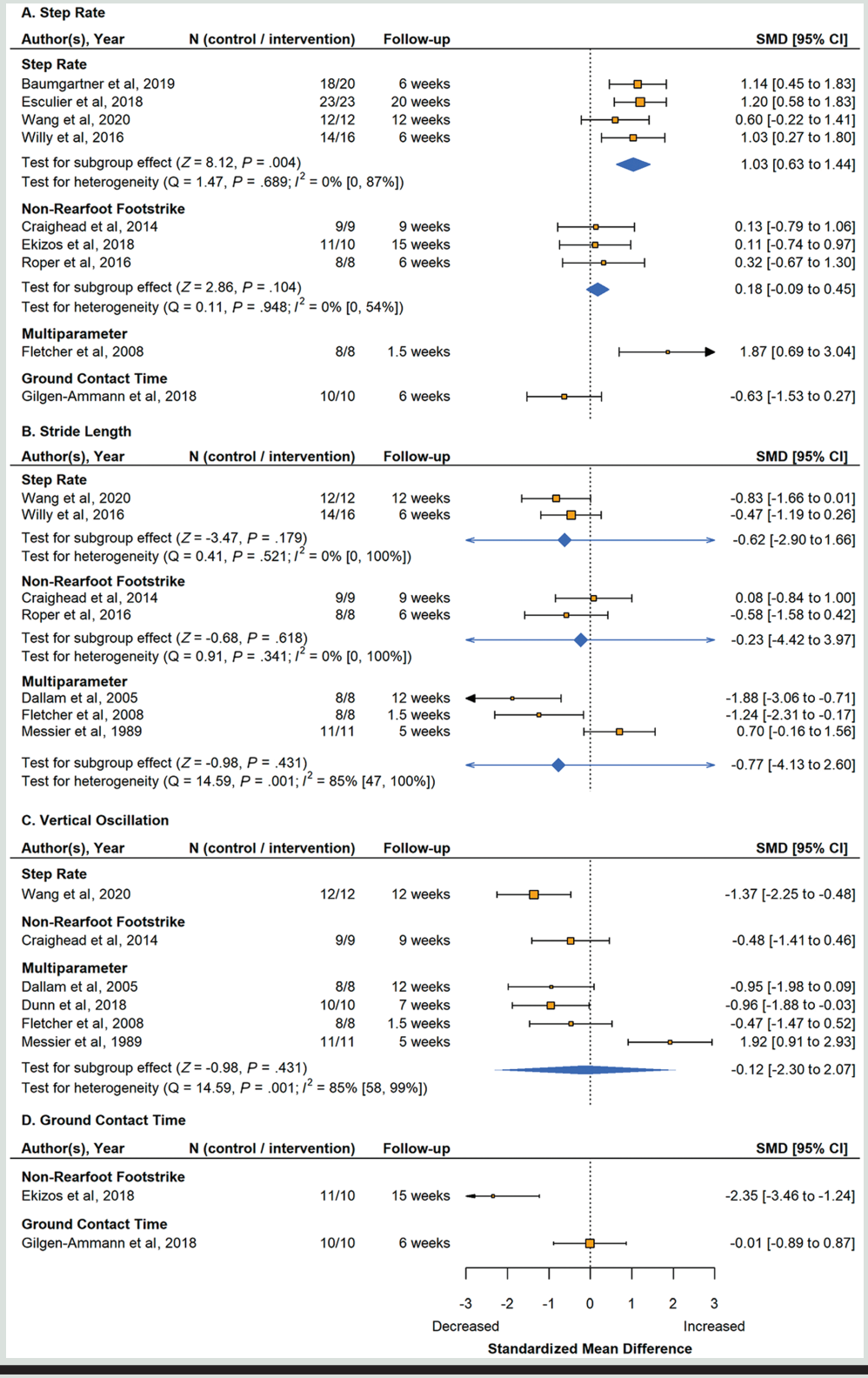

FIGURE 2. Spatiotemporal outcomes. Forest plot detailing standardized mean differences for spatiotemporal outcomes comparing runners who received gait retraining to control conditions. Abbreviations: CI, confidence interval; SMD, standardized mean difference.

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TABLE 2 Quality of Evidence Synthesis (GRADE)

		Summary of Findings			Certainty of Evidence			
Outcome Measure	**Subgroup**	***k***	**n**	**SMD (95% CI)**	**Risk of Bias**	**Imprecision**	**Inconsistency**	**Overall Certainty**
Spatiotemporal outcomes								
Step rate	Step rate	4	138	1.03 (0.63 to 1.44)	None	-1	None	Moderate
	Non-rearfoot footstrike	3	55	0.18 (-0.09 to 0.45)	-1*	-1	None	Low
Stride length	Step rate	2	54	-0.62 (-2.90 to 1.66)	None	-1	None	Moderate
	Non-rearfoot footstrike	2	34	-0.23 (-4.42 to 3.97)	-1*	-1	None	Low
	Multiparameter	3	54	-0.77 (-4.13 to2.60)	-1*	-1	-1	Very low
Vertical oscillation	Multiparameter	4	74	-0.12 (-2.30 to2.07)	-1*	-1	-1	Very low
Kinematic outcomes								
Ankle dorsiflexion at initial contact	Non-rearfoot footstrike	2	34	-0.87 (-2.54 to 0.80)	-1*	-1	-1	Very low
Knee flexion angle at initial contact	Non-rearfoot footstrike	2	34	0.74 (0.11 to 1.37)	-1*	-1	None	Low
Peak knee flexion angle during stance	Non-rearfoot footstrike	2	34	-0.19 (-4.70 to 4.32)	-1*	-1	None	Low
	Multiparameter	2	42	0.44 (-11.32 to 12.20)	None	-1	-1	Low
Peak hip adduction angle during stance	Step rate	2	76	-0.34 (-2.74 to 2.05)	None	-1	None	Moderate
Footstrike angle at initial contact	Step rate	2	54	0.76 (-2.96 to 1.43)	None	-1	None	Moderate
Kinetic outcomes								
Average vertical loading rate	Step rate	3	100	-0.57 (-1.05 to -0.09)	None	-1	None	Moderate
	Impact	2	342	-0.99 (-8.61 to 6.64)	None	-1	-1	Low
	Multiparameter	2	44	-0.55 (-5.63 to 4.52)	-1*	-1	None	Low
Instantaneous vertical loading rate	Step rate	2	54	-0.76 (-2.32 to 0.79)	None	-1	None	Moderate
	Impact	2	342	-1.01 (-4.16 to 2.13)	None	-1	None	Moderate
Performance outcomes								
Timed run	Multiparameter	2	36	-0.14 (-4.87 to 4.58)	-1*	-1	None	Low
Oxygen consumption (steady state)	Non-rearfoot footstrike	3	55	0.21 (-1.11 to 1.52)	-1*	-1	None	Low
	Multiparameter	3	54	0.32 (-0.39 to 1.02)	-1*	-1	None	Low
Rate of perceived exertion	Multiparameter	3	38	0.11 (-5.23 to 5.45)	None	-1	None	Moderate

Abbreviations: CI, confidence interval; GRADE, Grading of Recommendations Assessment, Development and Evaluation; k, number of independent trials; SMD, standardized mean difference.
** > 50% of trials with > 1 risk of bias assessment item rated as high risk.*

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decrease in peak tibial acceleration (MD -1.4 g [95% CI, -2.9 to 0.2]).

Vertical Impact Peak Three trials,[9,30,48] from the step rate, impact, and multiparameter subgroups, reported on vertical impact peak. The trial using a step rate intervention[48] found a small significant decrease in vertical impact peak (MD -0.23 BW [95% CI, -0.43 to -0.03], whereas trivial to small changes were found in trials using impact[9] (MD 0.05 BW [95% CI, -0.09 to 0.19]) and multiparameter [30] (MD 0.03 [95% CI, -0.56 to 0.62]) interventions.

Patellofemoral Joint Reaction Force Two trials,[19,43] from the step rate and nonrearfoot footstrike subgroups, found a moderate (MD -0.40 BW [95% CI, -0.78 to -0.02]) and a large (MD -1.31 BW [95% CI, -2.25 to -0.37]) significant reduction in peak patellofemoral joint reaction force following gait retraining, respectively.

Performance Outcomes

Nine trials[9,10,13,17,18,21,22,35,42] measured outcomes relating to running performance following gait retraining. SMD pooled results and grades of evidence are presented in **FIGURE 5** and **TABLE 2**, whereas corresponding MDs are reported in this subsection text. MDs could not be calculated for timed run or running economy outcomes based on differences in units of measurement.

Timed Run Low-certainty evidence from 2 trials[17,21] indicates that multiparameter interventions have a trivial nonsignificant effect on timed run performance (**FIGURE 5**).

Steady-State Oxygen Consumption (Running Economy) There was low-certainty

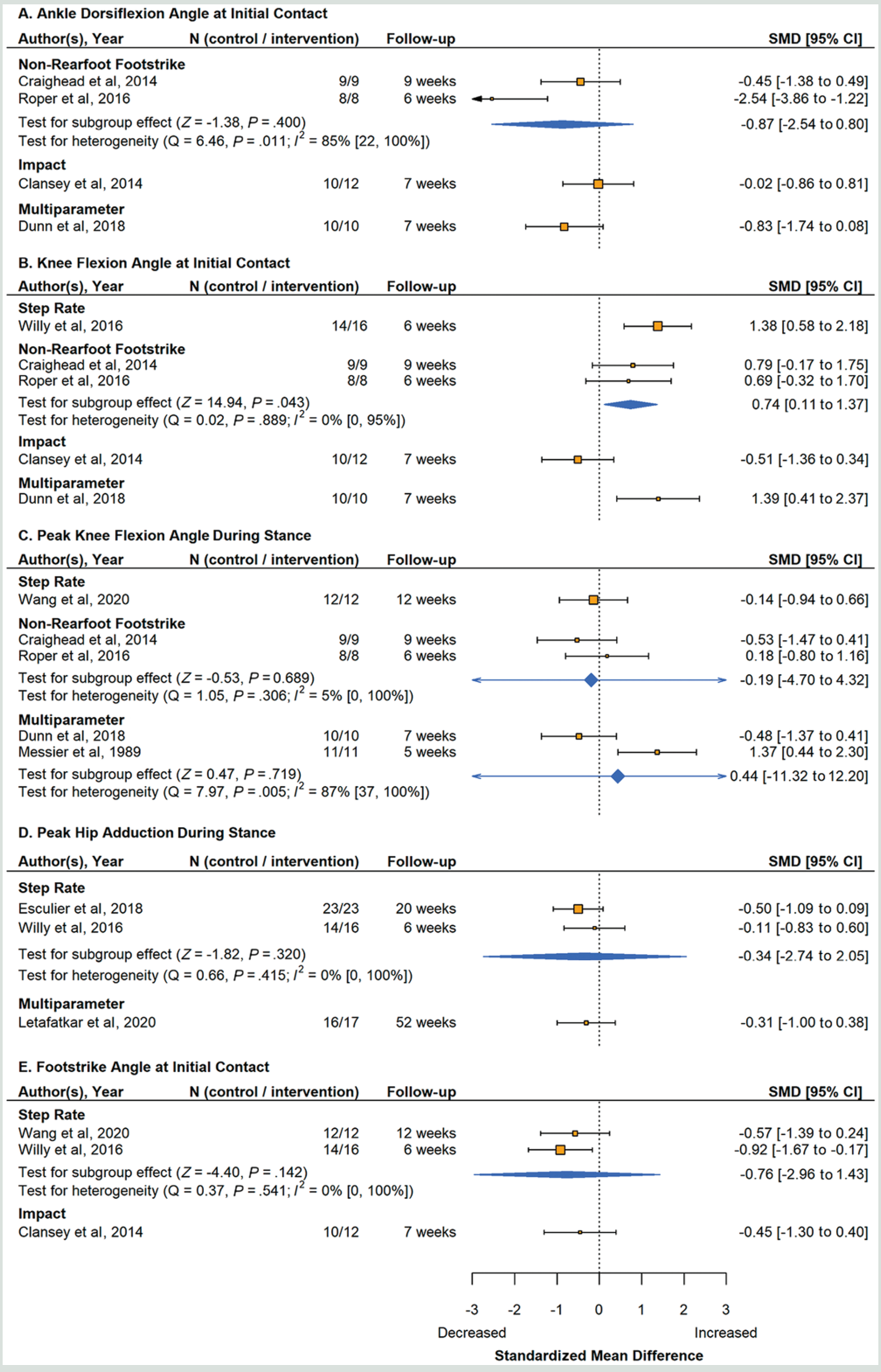

FIGURE 3. Kinematic outcomes. Forest plot detailing standardized mean differences for running kinematics comparing runners who received gait retraining to control conditions. Footstrike angle refers to the sagittal plane angle of the foot relative to the ground. Abbreviations: CI, confidence interval; SMD, standardized mean difference.

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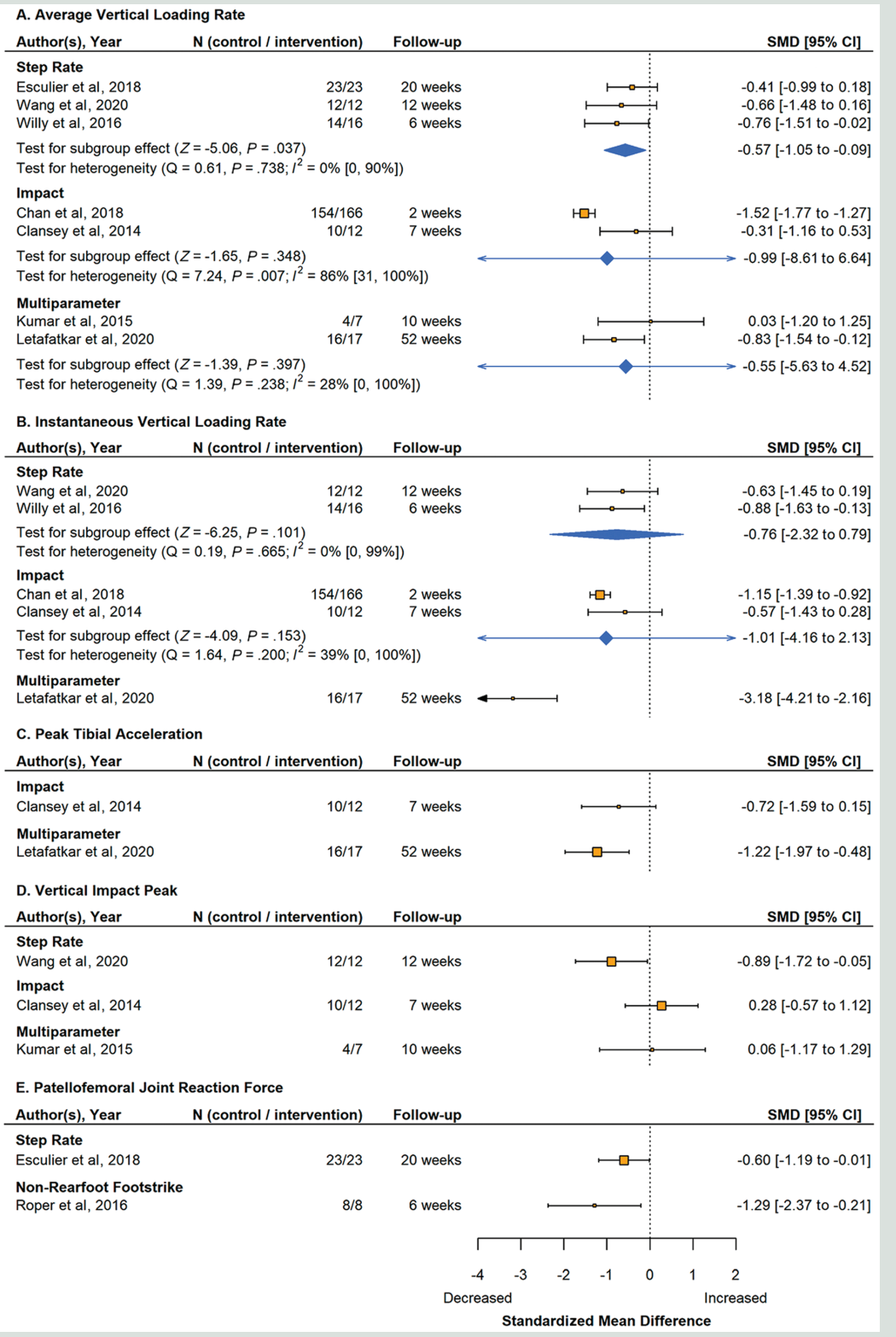

FIGURE 4. Kinetic outcomes. Forest plot detailing standardized mean differences for running kinetics comparing runners who received gait retraining to control conditions. Abbreviations: CI, confidence interval; SMD, standardized mean difference.

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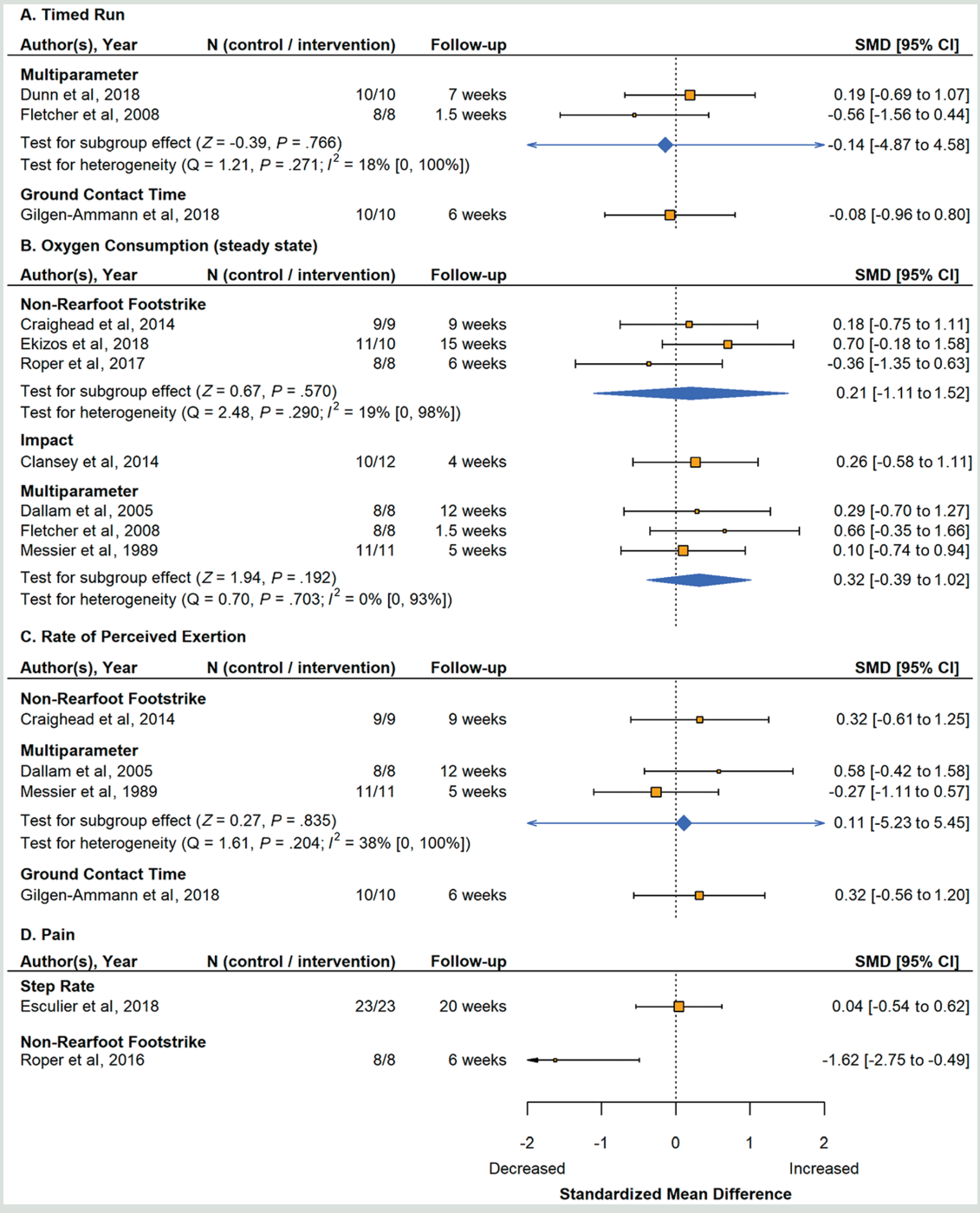

FIGURE 5. Clinical outcomes. Forest plot detailing standardized mean differences for running performance and pain outcomes comparing runners who received gait retraining to control conditions. Abbreviations: CI, confidence interval; SMD, standardized mean difference.

evidence from 3 trials[10,18,43] that nonrearfoot footstrike retraining has only trivial nonsignificant effects on running economy (**FIGURE 5**). Low-certainty evidence from 3 trials[13,21,35] demonstrated that multiparameter interventions produce a small nonsignificant increase in oxygen consumption (**FIGURE 5**).

Rate of Perceived Exertion There was moderate-certainty evidence from 2 trials[13,35] that multiparameter interventions do not affect rate of perceived exertion

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with only trivial nonsignificant changes observed (MD 0.3 [95% CI, -11.7 to 12.4], Borg scale, 6-20).

Clinical Outcomes

Two trials[19,43] examined the effects on pain, and 2 trials[8,32] reported injury outcomes in runners following gait retraining. Individual trial results for pain outcomes are presented in **FIGURE 5**, whereas corresponding MDs are reported in this subsection text.

Pain Two trials[19,43] examined the effects of gait retraining on pain in runners, both including participants with patellofemoral pain, with conflicting results. One trial,[43] including 16 runners, investigated whether a non-rearfoot footstrike intervention reduces knee pain. The retraining intervention consisted of 8 sessions with faded feedback focused on changing from a rearfoot to forefoot footstrike, whereas the control group performed identical sessions without feedback. A large significant decrease in knee pain in runners who received gait retraining was observed at 1 month (MD -2.7 [95% CI, -4.2 to -1.2], visual analog scale, 0-10). In contrast, the other trial[19] reported no additional benefits of gait retraining and patient education compared to education alone in 69 runners with patellofemoral pain (MD 0.1 [95% CI, -1.3 to 1.5], visual analog scale, 0-10). The retraining intervention involved 5 × 10-minute sessions, primarily focused on increasing step rate by 7.5%-10%, combined with patient education on load management.

Injury Two trials[8,32] examining the effects of gait retraining on injury outcomes demonstrated a significant reduction in 1-year injury incidence. One trial,[8] including 320 runners, examined the effects of 8 sessions of real-time visual feedback of impact forces. There was a 62% lower injury risk compared to no feedback (hazard ratio 0.38 [95% CI, 0.25 to 0.59]), based on the time-to-injury analysis. The remaining smaller trial,[32] including 49 male runners, examined the effects of an 8-week, 24-session conditioning program, with and without the inclusion of multiparameter-based gait retraining. Runners who did receive gait retraining had a 48% lower injury risk than runners who did not receive feedback, based on analysis of injury count data (rate ratio 0.52 [95% CI, 0.31 to 0.86].[32]

Adverse Events Two trials[9,43] provided information on adverse events following gait retraining. The 2 trials using impact[9] and non-rearfoot footstrike[43] interventions reported general calf soreness in the retraining intervention groups. However, they did not provide the incidence of soreness or any comparative information from the control groups. Additionally, 2 incidences of ankle soreness were reported in the intervention group in 1 trial[43] using a non-rearfoot footstrike intervention.

DISCUSSION

We found moderate-certainty evidence that step rate interventions cause a moderate increase in step rate and a small decrease in average vertical loading rate. We found low-certainty evidence that non-rearfoot footstrike interventions result in a moderate increase in knee flexion angle at initial contact, and non-rearfoot footstrike and multiparameter interventions do not affect running economy or performance. Few trials investigated the effects of gait retraining on pain and injury outcomes. The limited evidence suggested unclear effects on knee pain but possible injury prevention benefits of multiparameter and impact-based gait retraining.

Effects of Gait Retraining on Spatiotemporal and Kinematic Variables

We found moderate-certainty evidence of increased step rate with step rate–based interventions and low-certainty evidence that non-rearfoot footstrike interventions do not change step rate. Increasing step rate is a common biomechanical goal of gait retraining,[15,16,24] and it has been suggested as an intervention for injured runners in a previous review[4] and in recent patellofemoral pain clinical practice guidelines.[51] Clinicians and coaches should use feedback that directly targets step rate to accomplish a change in step rate.

Non-rearfoot footstrike retraining was the only intervention that changed joint kinematics during running. Unexpectedly, there was no change in ankle kinematics, but there was an increase in knee flexion at initial contact with non-rearfoot footstrike retraining. The increase in knee flexion at initial contact may account for the lower overall knee flexion range of motion observed with non-rearfoot footstrike interventions in a recent review of uncontrolled studies.[2] The lack of ankle kinematic changes is inconsistent with a review by Napier *et al*,[38] which reported large changes in sagittal joint ankle kinematics at initial contact with a midfoot or forefoot footstrike, primarily based on cross-sectional studies. Our null finding is based on longitudinal follow-up after non-rearfoot footstrike intervention and implies that longer-term effects on ankle kinematics are different from the acute changes reported previously.

Effects of Gait Retraining on Running Kinetics

Impact- and step rate–based gait retraining demonstrated moderate and small reductions on vertical loading rate variables, respectively. These findings were supported by low- to moderate-certainty evidence. However, the step rate–based effects may not be clinically relevant due to the small effect size. The impact-based effects were not statistically significant and uncertain due to the high variance of trial sample sizes and moderate heterogeneity between trials. Previous reviews[1,44] containing non-RCT evidence and several case series[7,39,45] have demonstrated reduced loading rates with impact-based gait retraining. Our review of RCTs highlights that this effect is uncertain when considering only the highest available evidence, and practitioners should interpret with caution until further RCTs are completed.

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Trials reporting on peak tibial acceleration and peak patellofemoral joint reaction force were too different to pool data. Results from individual trials demonstrated reductions in these outcomes across multiple retraining interventions. These findings were consistent with nonrandomized crossover[7,45] and noncontrolled intervention[11] studies on peak tibial acceleration and cross-sectional studies on patellofemoral pain.[31,49] These reductions may be clinically meaningful as elevated peak tibial acceleration and patellofemoral joint reaction forces have been associated with lower limb stress fractures[36] and patellofemoral pain,[40] respectively.

Effects of Gait Retraining on Running Performance

We found low- to moderate-certainty evidence that gait retraining, using nonrearfoot footstrike or multiparameter interventions, does not improve or hinder running economy or performance. A previous review[2] reported that transitioning from a rearfoot to a non-rearfoot footstrike (medium speeds) reduces running economy in the immediate term. Many coaches and researchers believe runners naturally adopt the most economical running pattern, and changing this pattern increases metabolic demands.[34,37] Our null finding suggests that clinicians, coaches, and runners need not be concerned about gait retraining having (longer-term) detrimental effects on running economy and performance.

Effects of Gait Retraining on Clinical Outcomes

Patellofemoral pain[19,43] and lower limb injury[8,32] were each investigated by only 2 trials using different gait retraining strategies. Further research is needed to quantify the extent to which runners and practitioners can be confident in the injury prevention benefits reported by individual gait retraining trials. Greater available evidence would also help determine why some studies report beneficial effects of gait retraining on pain outcomes,[6,43] and others report no effects.[19]

Clinical Implications

Gait retraining interventions should target the desired kinematic or kinetic outcomes of interest to produce the most meaningful effects. The gait retraining approaches in our review do not appear to have any lasting negative effects on running performance or economy but equally cannot be recommended as a method to improve running performance. There is insufficient evidence to recommend gait retraining for injury prevention and treatment at this stage.

Research Implications

The certainty of evidence was generally low to moderate due to imprecision related to the small size of most trials and high risk of bias. To improve evidence quality, researchers should ensure random allocation of participants, ensure appropriate blinding of outcome assessors, and assess retention of retrained gait patterns without participant awareness of sampling timing. We recommend that additional RCT evidence is required to better inform gait retraining design and implementation, with a particular focus on injured populations who are most commonly targeted for gait retraining by practitioners.

Limitations

There was large heterogeneity in gait retraining interventions across the included trials that limited data pooling and led to wide CIs and reduced the certainty of findings. Although we categorized trials into subgroups based on the retraining intervention for meta-analysis purposes, there was still some variance within subgroups, particularly for the multiparameter interventions that used different combinations of feedback strategies between trials.

There was a diverse range of follow-up durations across included trials. We extracted data from the longest follow-up duration to best indicate the longer-term effect of gait retraining. Choosing a long follow-up duration may have led to our findings providing a lesser effect of gait-retraining based on the potential for runners to revert to the pretraining running technique.[9] However, we believe that describing the lasting effects of gait retraining is more insightful for runners and practitioners.

Finally, not all trials in our review selectively recruited runners with known biomechanical deviations (eg, high impact forces). The effects of gait retraining might be larger if studies recruit participants according to the biomechanical outcomes they wish to retrain.

CONCLUSION

There was moderate-certainty evidence that step rate–based gait retraining increased step rate and reduced average vertical loading rate and low-certainty evidence that non-rearfoot footstrike retraining increased knee flexion at initial contact. Low-certainty evidence suggested that running economy and performance measures were unaffected by gait retraining using multiparameter or non-rearfoot footstrike interventions. Insufficient trials were available to determine the effects of gait retraining on pain and injury. ◉

KEY POINTS

FINDINGS: Moderate-certainty evidence indicated that step rate–based gait retraining increased step rate and reduced average vertical loading rate. Low-certainty evidence indicated that running economy or performance measures were unaffected by gait retraining approaches using multiparameter or non-rearfoot footstrike interventions. Insufficient trials reported on pain and injury outcomes, with limited evidence from individual trials indicating varied findings for pain but promising preliminary findings for injury prevention.

IMPLICATIONS: Clinicians and researchers can selectively use the gait retraining approaches described in this review to target specific kinematic or kinetic changes but not enhance running performance. Gait retraining targeting a reduction in landing impacts may reduce

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injury incidence in healthy runners, but the effect on pain in injured runners is unclear.

CAUTION: Further research involving injured runners is required because they are the population considered most in need of retraining. Gait retraining may reduce injury incidence, but further research is needed to validate this finding before being recommended as an injury prevention strategy.

STUDY DETAILS

AUTHOR CONTRIBUTIONS: All authors contributed to the conception and design of the review and completion of the search strategy. Eoin Doyle and Joel Fuller were responsible for the meta-analysis. Eoin Doyle drafted the manuscript. All authors edited and revised the manuscript and approved the final version of the manuscript.

DATA SHARING: All data have been extracted from the included articles. Data used for meta-analysis can be found in **SUPPLEMENTAL MATERIAL**.

PATIENT AND PUBLIC: There was no patient, public, or athlete involvement in this study.

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REFERENCES

1. Agresta C, Brown A. Gait retraining for injured and healthy runners using augmented feedback: A systematic literature review. *J Orthop Sports Phys Ther.* 2015;45:576-584. https://doi.org/10.2519/jospt.2015.5823
2. Anderson LM, Bonanno DR, Hart HF, Barton CJ. What are the benefits and risks associated with changing foot strike pattern during running? A systematic review and meta-analysis of injury, running economy, and biomechanics. *Sports Med.* 2020;50:885-917. https://doi.org/10.1007/s40279-019-01238-y
3. Atkins D, Best D, Briss PA, et al. Grading quality of evidence and strength of recommendations. *BMJ.* 2004;328:1490. https://doi.org/10.1136/bmj.328.7454.1490
4. Barton CJ, Bonanno DR, Carr J, et al. Running retraining to treat lower limb injuries: A mixed-methods study of current evidence synthesised with expert opinion. *Br J Sports Med.* 2016;50:513-526. https://doi.org/10.1136/bjsports-2015-095278
5. Baumgartner J, Gusmer R, Hollman J, Finnoff JT. Increased stride-rate in runners following an independent retraining program: A randomized controlled trial. *Scand J Med Sci Sports.* 2019;29:1789-1796. https://doi.org/10.1111/sms.13509
6. Bonacci J, Hall M, Saunders N, Vicenzino B. Gait retraining versus foot orthoses for patellofemoral pain: A pilot randomised clinical trial. *J Sci Med Sport.* 2018;21:457-461. https://doi.org/10.1016/j.jsams.2017.09.187
7. Bowser BJ, Fellin R, Milner CE, Pohl MB, Davis IS. Reducing impact loading in runners: A one-year follow-up. *Med Sci Sports Exerc.* 2018;50:2500-2506. https://doi.org/10.1249/MSS.0000000000001710
8. Chan ZYS, Zhang JH, Au IPH, et al. Gait Retraining for the Reduction of Injury Occurrence in Novice Distance Runners: 1-Year Follow-up of a Randomized Controlled Trial. *Am J Sports Med.* 2018;46:388-395. https://doi.org/10.1177/0363546517736277
9. Clansey AC, Hanlon M, Wallace ES, Nevill A, Lake MJ. Influence of tibial shock feedback training on impact loading and running economy. *Med Sci Sports Exerc.* 2014;46:973-981. https://doi.org/10.1249/MSS.0000000000000182
10. Craighead DH, Lehecka N, King DL. A novel running mechanic's class changes kinematics but not running economy. *J Strength Cond Res.* 2014;28:3137-3145. https://doi.org/10.1519/JSC.0000000000000500
11. Crowell HP, Davis IS. Gait retraining to reduce lower extremity loading in runners. *Clin Biomech (Bristol, Avon).* 2011;26:78-83. https://doi.org/10.1016/j.clinbiomech.2010.09.003
12. Crowell HP, Milner CE, Hamill J, Davis IS. Reducing impact loading during running with the use of real-time visual feedback. *J Orthop Sports Phys Ther.* 2010;40:206-213. https://doi.org/10.2519/jospt.2010.3166
13. Dallam GM, Wilber RL, Jadelis K, Fletcher G, Romanov N. Effect of a global alteration of running technique on kinematics and economy. *J Sports Sci.* 2005;23:757-764. https://doi.org/10.1080/02640410400022003
14. Davis I. Optimising the efficacy of gait retraining. *Br J Sports Med.* 2018;52:624-625. https://doi.org/10.1136/bjsports-2017-098297
15. Davis IS, Futrell E. Gait retraining: Altering the fingerprint of gait. *Phys Med Rehabil Clin N Am.* 2016;27:339-355. https://doi.org/10.1016/j.pmr.2015.09.002
16. Davis IS, Tenforde AS, Neal BS, Roper JL, Willy RW. Gait retraining as an intervention for patellofemoral pain. *Curr Rev Musculoskelet Med.* 2020;13:103-114. https://doi.org/10.1007/s12178-020-09605-3
17. Dunn MD, Claxton DB, Fletcher G, Wheat JS, Binney DM. Effects of running retraining on biomechanical factors associated with lower limb injury. *Hum Mov Sci.* 2018;58:21-31. https://doi.org/10.1016/j.humov.2018.01.001
18. Ekizos A, Santuz A, Arampatzis A. Short- and long-term effects of altered point of ground reaction force application on human running energetics. *J Exp Biol.* 2018;221:jeb176719. https://doi.org/10.1242/jeb.176719
19. Esculier JF, Bouyer LJ, Dubois B, et al. Is combining gait retraining or an exercise programme with education better than education alone in treating runners with patellofemoral pain? A randomised clinical trial. *Br J Sports Med.* 2018;52:659-666. https://doi.org/10.1136/bjsports-2016-096988
20. Esculier JF, Bouyer LJ, Dubois B, et al. Predictors of clinical success in runners with patellofemoral pain: Secondary analyses of a randomized clinical trial. *J Sci Med Sport.* 2018;21:777-782. https://doi.org/10.1016/j.jsams.2018.01.006
21. Fletcher G, Bartlett R, Romanov N, Fotouhi A. Pose (R) method technique improves running performance without economy changes. *Int J Sports Sci Coach.* 2008;3:365-380. https://doi.org/10.1260/174795408786238506
22. Gilgen-Ammann R, Wyss T, Troesch S, Heyer L, Taube W. Positive effects of augmented feedback to reduce time on ground in well-trained runners. *Int J Sports Physiol Perform.* 2018;13:88-94. https://doi.org/10.1123/ijspp.2016-0746
23. Hafer JF, Brown AM, deMille P, Hillstrom HJ, Garber CE. The effect of a cadence retraining protocol on running biomechanics and efficiency: A pilot study. *J Sports Sci.* 2015;33:724-731. https://doi.org/10.1080/02640414.2014.962573
24. Heiderscheit BC. Gait retraining for runners: In search of the ideal. *J Orthop Sports Phys Ther.* 2011;41:909-910. https://doi.org/10.2519/jospt.2011.0111
25. Higgins JP, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ.* 2011;343:d5928. https://doi.org/10.1136/bmj.d5928.
26. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ.* 2003;327:557-560. https://doi.org/10.1136/bmj.327.7414.557
27. Hopkins WG. A new view of statistics. Internet Society for Sport Science. Available at: http://www.sportsci.org/resource/stats/. Accessed April 5, 2020.
28. Inthout J, Ioannidis JP, Borm GF. The Hartung-Knapp-Sidik-Jonkman method for random effects meta-analysis is straightforward and considerably outperforms the standard DerSimonian-Laird method. *BMC Med Res Methodol.* 2014;14:25. https://doi.org/10.1186/1471-2288-14-25
29. Kluitenberg B, van Middelkoop M, Diercks R, van der Worp H. What are the differences in injury proportions between different populations of runners? A systematic review and meta-analysis. *Sports Med.* 2015;45:1143-1161. https://doi.org/10.1007/s40279-015-0331-x
30. Kumar D, McDermott K, Feng H, et al. Effects of form-focused training on running biomechanics: A pilot randomized trial in untrained individuals. *PM R.* 2015;7:814-822. https://doi.org/10.1016/j.pmrj.2015.01.010
31. Lenhart RL, Thelen DG, Wille CM, Chumanov ES, Heiderscheit BC. Increasing running step

rate reduces patellofemoral joint forces. *Med Sci Sports Exerc.* 2014;46:557-564. https://doi.org/10.1249/MSS.0b013e3182a78c3a

32. Letafatkar A, Rabiei P, Farivar N, Alamouti G. Long-term efficacy of conditioning training program combined with feedback on kinetics and kinematics in male runners. *Scand J Med Sci Sports.* 2020;30:429-441. https://doi.org/10.1111/sms.13587
33. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: Explanation and elaboration. *BMJ.* 2009;339:b2700. https://doi.org/10.1136/bmj.b2700
34. McCann DJ, Higginson BK. Training to maximize economy of motion in running gait. *Curr Sports Med Rep.* 2008;7:158-162. https://doi.org/10.1097/01.CSMR.0000319711.63793.84
35. Messier SP, Cirillo KJ. Effects of a verbal and visual feedback system on running technique, perceived exertion and running economy in female novice runners. *J Sports Sci.* 1989;7:113-126. https://doi.org/10.1080/02640418908729830
36. Milner CE, Ferber R, Pollard CD, Hamill J, Davis IS. Biomechanical factors associated with tibial stress fracture in female runners. *Med Sci Sports Exerc.* 2006;38:323-328. https://doi.org/10.1249/01.mss.0000183477.75808.92
37. Moore IS, Jones AM, Dixon SJ. Mechanisms for improved running economy in beginner runners. *Med Sci Sports Exerc.* 2012;44:1756-1763. https://doi.org/10.1249/MSS.0b013e318255a727
38. Napier C, Cochrane CK, Taunton JE, Hunt MA. Gait modifications to change lower extremity gait biomechanics in runners: A systematic review. *Br J Sports Med.* 2015;49:1382-1388. https://doi.org/10.1136/bjsports-2014-094393
39. Napier C, MacLean CL, Maurer J, Taunton JE, Hunt MA. Real-time biofeedback of performance to reduce braking forces associated with running-related injury: An exploratory study. *J Orthop Sports Phys Ther.* 2019;49:136-144. https://doi.org/10.2519/jospt.2019.8587
40. Powers CM, Bolgla LA, Callaghan MJ, Collins N, Sheehan FT. Patellofemoral pain: Proximal, distal, and local factors, 2nd International Research Retreat. *J Orthop Sports Phys Ther.* 2012;42:A1-54. https://doi.org/10.2519/jospt.2012.0301
41. Rohatgi A. Web Plot Digitiser v4.4. Available at https://apps.automeris.io/wpd/. Published 2020. Accessed April 30, 2021.
42. Roper JL, Doerfler D, Kravitz L, Dufek JS, Mermier C. Gait retraining from rearfoot strike to forefoot strike does not change running economy. *Int J Sports Med.* 2017;38:1076-1082. https://doi.org/10.1055/s-0043-110225
43. Roper JL, Harding EM, Doerfler D, et al. The effects of gait retraining in runners with patellofemoral pain: A randomized trial. *Clin Biomech (Bristol, Avon).* 2016;35:14-22. https://doi.org/10.1016/j.clinbiomech.2016.03.010
44. Schubert AG, Kempf J, Heiderscheit BC. Influence of stride frequency and length on running mechanics: A systematic review. *Sports Health.* 2014;6:210-217. https://doi.org/10.1177/1941738113508544
45. Sheerin KR, Reid D, Taylor D, Besier TF. The effectiveness of real-time haptic feedback gait retraining for reducing resultant tibial acceleration with runners. *Physical Therapy in Sport.* 2020;43:173-180. https://doi.org/10.1016/j.ptsp.2020.03.001
46. van Gent RN, Siem D, van Middelkoop M, van Os AG, Bierma-Zeinstra SM, Koes BW. Incidence and determinants of lower extremity running injuries in long distance runners: A systematic review. *Br J Sports Med.* 2007;41:469-480. https://doi.org/10.1136/bjsm.2006.033548
47. Videbaek S, Bueno AM, Nielsen RO, Rasmussen S. Incidence of running-related injuries per 1000 h of running in different types of runners: A systematic review and meta-analysis. *Sports Med.* 2015;45:1017-1026. https://doi.org/10.1007/s40279-015-0333-8
48. Wang J, Luo Z, Dai B, Fu W. Effects of 12-week cadence retraining on impact peak, load rates and lower extremity biomechanics in running. *PeerJ.* 2020;8:e9813-e9813. https://doi.org/10.7717/peerj.9813
49. Willson JD, Sharpee R, Meardon SA, Kernozek TW. Effects of step length on patellofemoral joint stress in female runners with and without patellofemoral pain. *Clin Biomech (Bristol, Avon).* 2014;29:243-247. https://doi.org/10.1016/j.clinbiomech.2013.12.016
50. Willy RW, Buchenic L, Rogacki K, Ackerman J, Schmidt A, Willson JD. In-field gait retraining and mobile monitoring to address running biomechanics associated with tibial stress fracture. *Scand J Med Sci Sports.* 2016;26:197-205. https://doi.org/10.1111/sms.12413
51. Willy RW, Hoglund LT, Barton CJ, et al. Patellofemoral pain. *J Orthop Sports Phys Ther.* 2019;49:CPG1-CPG95. https://doi.org/10.2519/jospt.2019.0302
52. Willy RW, Meardon SA, Schmidt A, Blaylock NR, Hadding SA, Willson JD. Changes in tibiofemoral contact forces during running in response to in-field gait retraining. *J Sports Sci.* 2016;34:1602-1611. https://doi.org/10.1080/02640414.2015.1125517
53. Willy RW, Scholz JP, Davis IS. Mirror gait retraining for the treatment of patellofemoral pain in female runners. *Clin Biomech (Bristol, Avon).* 2012;27:1045-1051. https://doi.org/10.1016/j.clinbiomech.2012.07.011

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APPENDIX A

Search Strategy (March 18, 2021)

Medline (Ovid Interface)
#1 Running/ or Jogging/ (20945)
#2 (runn* or jogg*).tw. (69247)
#3 or/1-2 (75098)
#4 Gait/ or Kinetics/ (518765)
#5 (cadence or kinetic* or kinematic*).tw. (422981)
#6 (pressure or acceleration or force or moment or impulse or EMG or electromyography).tw. (1224620)
#7 (loading rate* or impact peak* or tibial shock).tw. (6251)
#8 (step adj (width or frequency or rate or length)).tw. (4114)
#9 (stride adj (frequency or length or rate)).tw. (3626)
#10 (foot adj (strike or angle)).tw. (623)
#11 (run* adj (technique or style or mechanics or form or pattern)).tw. (555)
#12 or/4-11 (1964731)
#13 (retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust).tw. (7942019)
#14 and/3,12,13 (5954)
#15 limit 14 to (english language and humans) (3620)

CINAHL (EBSCO Interface)
S1. (MH "running+") OR (MH "jogging+")
S2. TI (runn* or jogg*) OR AB (runn* or jogg*)
S3. S1 OR S2
S4. (MM "Gait") OR (MM "Gait Analysis")
S5. TI (cadence OR kinetic or kinematic) OR AB (cadence OR kinetic or kinematic)
S6. TI (pressure or acceleration or force or moment or impulse or EMG or electromyography) OR AB (pressure or acceleration or force or moment or impulse or EMG or electromyography)
S7. TI ("load* rate*" or "impact peak*" or "tibial shock") OR AB ("load* rate*" or "impact peak*" or "tibial shock")
S8. TI (step) W1 (width or frequency or rate or length) OR AB (step) W1 (width or frequency or rate or length)
S9. TI (stride W1 (frequency or length or rate)) OR AB (stride W1 (frequency or length or rate))
S10. TI (foot W1 (strike or angle)) OR AB (foot W1 (strike or angle))
S11. TI (run* W1 (technique or style or mechanics or form or pattern)) OR AB (run* W1 (technique or style or mechanics or form or pattern))
S12. S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11
S13. TI (retrain* or retrain* or modif* or manip* or feedback or increas* or decreas* or alter or adjust) OR AB (retrain* or retrain* or modif* or manip* or feedback or increas* or decreas* or alter or adjust)
S14. S3 AND S12 AND S13
S15. S3 AND S12

PEDro
#1. runn*.ti.ab (485)

Embase (Ovid Interface)
1. running/ or jogging/
2. (runn* or jogg*).ti,ab.
3. or/1-2 (94824)
4. gait/ or kinetics/
5. (cadence or kinetic* or kinematic*).ti,ab.
6. (pressure or acceleration or force or moment or impulse or EMG or electromyography).ti,ab.
7. (loading rate* or impact peak* or tibial shock).ti,ab.
8. (step adj (width or frequency or rate or length)).ti,ab.
9. (stride adj (frequency or length or rate)).ti,ab.
10. (foot adj (strike or angle)).ti,ab.
11. (run* adj (technique or style or mechanics or form or pattern)).ti,ab.

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APPENDIX A (CONTINUED)

12. or/4-11
13. (retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust).ti,ab.
14. 3 and 12 and 13
15. limit 14 to (english language and humans)

SPORTDiscus (EBSCO Interface)

S1 (DE "RUNNING" OR DE "BAREFOOT running" OR DE "CROSS-country running" OR DE "GAIT in humans" OR DE "JOGGING" OR DE "LONG-distance running" OR DE "MIDDLE distance running" OR DE "MINIMALIST running" OR DE "ROAD running" OR DE "RUNNING for women" OR DE "TRAIL running")
S2 TI (runn* or jogg*) OR AB (runn* or jogg*)
S3 S1 OR S2
S4 DE "GAIT in humans" or DE "kinematics"
S5 TI (cadence OR kinetic* or kinematic*) or AB (cadence OR kinetic* or kinematic*)
S6 TI (step W1 (width or frequency or rate or length or kinematics)) OR AB (step W1 (width or frequency or rate or length or kinematics))
S7 TI (stride W1 (frequency or length or rate)) OR AB (stride W1 (frequency or length or rate))
S8 TI (foot W1 (strike or angle)) OR AB (foot W1 (strike or angle))
S9 TI ("load* rate*" OR "impact peak*" OR "tibial shock") OR AB ("load* rate*" OR "impact peak*" OR "tibial shock")
S10 TI (pressure or acceleration or force or moment or impulse or EMG or electromyography) OR AB (pressure or acceleration or force or moment or impulse or EMG or electromyography)
S11 TI (run* W1 (technique or style or mechanics or form or pattern)) OR AB (run* W1 (technique or style or mechanics or form or pattern))
S12 S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11
S13 S3 AND S12 Limiters - Language: English
S14 TI (retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust) OR AB (retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust) Limiters - Language: English
S15 S13 AND S14 Limiters - Language: English

Web of Science (Web of Science Core Collection)

#1 TS=(runn* OR jogg*)
#2 TS=(gait)
#3 TI=(cadence or kinetic* or kinematic*) OR AB=(cadence or kinetic* or kinematic*)
#4 TI=(force or moment or impulse or EMG or electromyography) OR AB=(force or moment or impulse or EMG or electromyography)
#5 TI=("loading rate*" or "impact peak*" or "tibial shock") OR AB=("loading rate*" or "impact peak*" or "tibial shock")
#6 TI=("step width" OR "step frequency" or "step rate" or "step length") OR AB=("step width" OR "step frequency" or "step rate" or "step length")
#7 TI=("stride frequency" or "stride length" or "stride rate" OR "foot strike" or "foot angle") OR AB=("stride frequency" or "stride length" or "stride rate" OR "foot strike" or
"foot angle")
#8 TI=("foot strike" or "foot angle") OR AB=("foot strike" or "foot angle")
#9 TI=("runn* technique or "runn* style or "run* mechanics" or "run* form" or "run* pattern") OR AB=("runn* technique or "runn* style or "run* mechanics" or "run* form" or
"run* pattern")
#10 #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9#11 TI=(retrain* or re-train*
or modif* or manip* or feedback or increas* or decreas* or alter or adjust) OR AB=(retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust)
#12 (#1 AND #10 AND #11)
#13 (#1 AND #10 AND #11) AND LANGUAGE: (English)
Refined by: WEB OF SCIENCE CATEGORIES: (SPORT SCIENCES OR ORTHOPEDICS OR PHYSIOLOGY OR ENGINEERING BIOMEDICAL OR BIOPHYSICS OR MULTIDISCIPLINARY SCIENCES) AND WEB OF SCIENCE CATEGORIES: (SPORT SCIENCES OR ORTHOPEDICS OR PHYSIOLOGY OR ENGINEERING BIOMEDICAL OR BIOPHYSICS OR MULTIDISCIPLINARY SCIENCES OR REHABILITATION)

CENTRAL (Cochrane Library Interface)

#1 MeSH descriptor: [Running] explode all trees 2025
#2 MeSH descriptor: [Jogging] explode all trees 51
#3 (runn*):ti,ab,kw7297
#4 (jogg*):ti,ab,kw 513

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APPENDIX A (CONTINUED)

#5	{OR #1-#4}	7676
#6	MeSH descriptor: [Gait] explode all trees	2066
#7	MeSH descriptor: [Gait Analysis] explode all trees	31
#8	MeSH descriptor: [Biomechanical Phenomena] explode all trees	2779
#9	(cadence* or kinetic* or kinematic):ti,ab,kw	12765
#10	(pressure or acceleration or force or moment or impulse or EMG or electromyography):ti,ab,kw	172229
#11	((loading NEXT rate*) or (impact NEXT peak*) or "tibial shock"):ti,ab,kw	85
#12	(step NEXT (width or frequency or rate or length)):ti,ab,kw	770
#13	(stride NEXT (frequency or length or rate)):ti,ab,kw	729
#14	(foot NEXT (strike or angle)):ti,ab,kw	64
#15	(run* NEXT (technique or style or mechanics or form or pattern)):ti,ab,kw	91
#16	{OR #6-#15}	185674
#17	(retrain* or re-train* or modif* or manip* or feedback or increas* or decreas* or alter or adjust):ti,ab,kw	585633
#18	{AND #5, #16, #17}	1014
#19	#18 in Trials	992

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APPENDIX B

Statistical Code (RStudio)

```
#This line installs the required package for performing meta-analysis.
install.packages("metafor")
#This line switches the downloaded package.
library(metafor)
#This line reads in the txt data file.
mydata_SR <- read.table("Running_SR.txt",header=TRUE, sep="\t")
#This line calculates SMD and variance for subsequent meta-analysis (includes Hedges' correction).
ES_SR <- escalc(measure="SMD", m1i=Mean_Int,m2i=Mean_Con, sd1i=SD_Int, sd2i=SD_Con, n1i=N_Int,n2i=N_Con, data=mydata_SR)
#This line performs the subgroup meta-analyses with Hartung-Knapp adjustment, prints the summary output as well as calculating and displaying heterogeneity statistics. This subset option within rma was used to subgroup the analyses across intervention categories; step rate (Subgroup 1), non-rearfoot footstrike (Subgroup 2), multiparameter (Subgroup 3), and impact (Subgroup 4). Ground contact time was not used as an independent subgroup for meta-analysis.
meta.SR_SR <- rma(yi,vi, method="REML", test="knha",data=ES_SR,subset = (Subgroup==1))
SR_SR_I2_CI <- confint(meta.SR_SR)
summary(meta.SR_SR)
SR_SR_I2_CI
meta.SR_FS <- rma(yi,vi, method="REML", test="knha",data=ES_SR,subset = (Subgroup==2))
SR_FS_I2_CI <- confint(meta.SR_FS)
summary(meta.SR_FS)
meta.SR_multi <- rma(yi,vi,method="REML", test="knha",data=ES_SR,subset = (Subgroup==3))
SR_multi_I2_CI <- confint(meta.SR_multi)
summary(meta.SR_multi)
```

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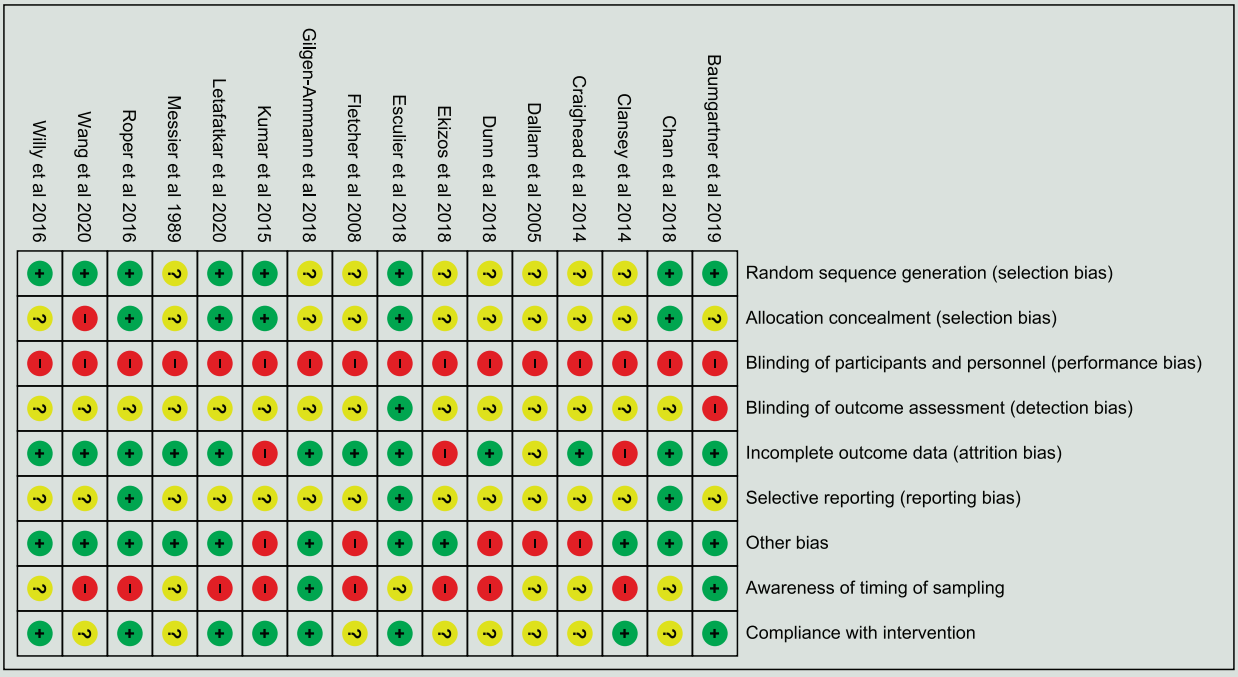
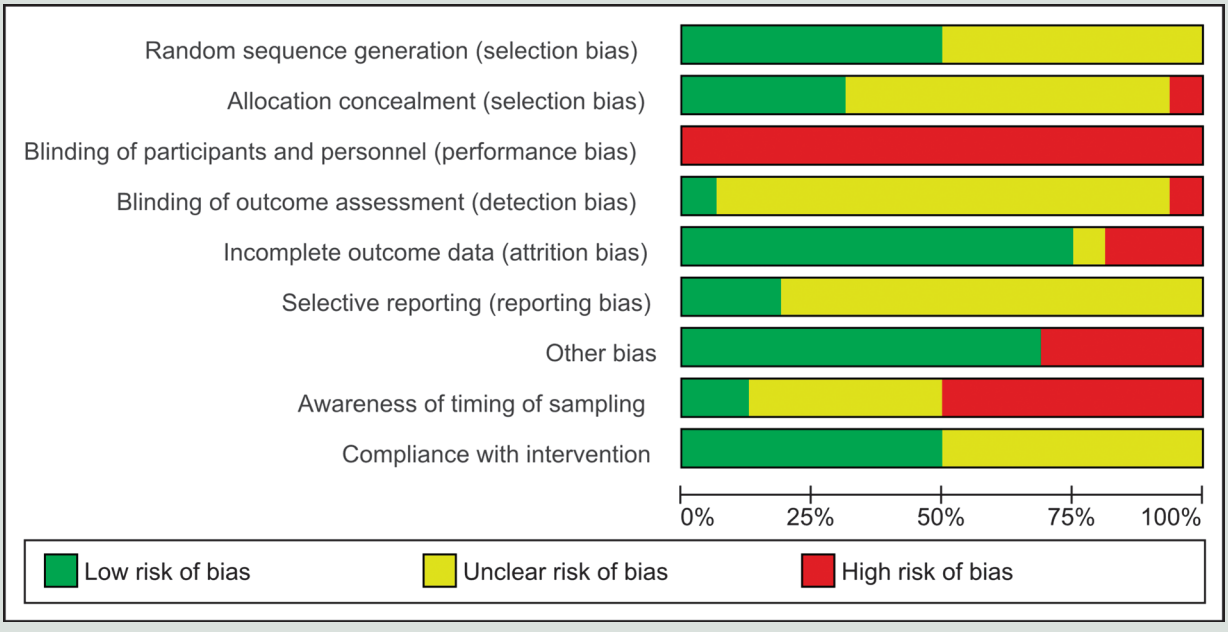
Risk of Bias Summary (A) and Graph (B)

A. Risk of bias summary: Review authors' judgments about each risk of bias item for each included study.



B. Risk of bias graph: Review authors' judgments about each risk of bias item presented as percentages across all included studies.



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